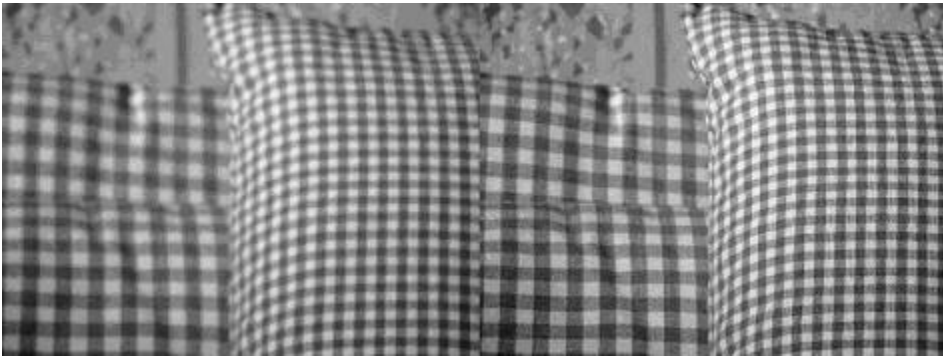


Depth Data Acquisition Using the Defocusing Method

This exercise is based on the work done by Akihiro Horii, The Royal Institute of Technology, Sweden.



A pair of differently focused images is used to acquire depth data.



The relative depth map indicates the structure of the scene.

The laboratory exercise time reservation list is at TE407. Select one of the times in groups of two. The exercise will be 16.2.-2.3.2006. Return the code early enough to reserve yourself time for making the corrections.

There are **12 pre-questions** that you have to answer **before** the exercise at the laboratory.

Preliminary questions

The goal of these 12 questions is to get us familiar with some concepts that are frequently used in photography and microscopy.

Explain briefly:

1) Focal length & aperture

http://www.dpreview.com/learn/?/Glossary/Optical/Focal_Length_01.htm

2) F-number of a lens system

3) Depth of field & depth of focus <http://photographytips.com/page.cfm/1601>

http://www.dpreview.com/learn/?/Glossary/Optical/Depth_of_Field_01.htm

4) Pixels vs. effective pixels

http://www.dpreview.com/learn/?/Glossary/Camera_System/Effective_Pixels_01.htm

5) Microscope optical components: how is the specimen magnified?

<http://www.olympusmicro.com/primer/anatomy/magnification.html>

What is empty magnification?

6) Image formation artifacts

http://www.dpreview.com/learn/?/Glossary/Digital_Imaging/Artifacts_01.htm

Answer the questions:

7) We want to maximise microscope image resolution. The goal is a total magnification of 250. Should we select a) 25x-eyepiece & 10x-objective or b) 10x-eyepiece & 25x-objective?

<http://www.olympusmicro.com/primer/anatomy/numaperture.html>

<http://www.olympusmicro.com/primer/java/nuaperture/index.html>

8) We want to maximise image sharpness (depth of field) with respect to the distance from the camera. Focal length is fixed. Name one camera adjustment to be made.

9) Image resolution is neither the number of pixels on the sensor

nor the size of the output image. What is image resolution?

http://www.dpreview.com/learn/?/Glossary/Digital_Imaging/Resolution_01.htm

10) What are the causes of the following image formation artifacts?

- a) Pincushion & barrel distortion
- b) Chromatic aberration
- c) Vignetting

http://www.dpreview.com/learn/?/Glossary/Optical/Pincushion_Distortion_01.htm

http://www.dpreview.com/learn/?/Glossary/Optical/Barrel_Distortion_01.htm

http://www.dpreview.com/learn/?/Glossary/Optical/chromatic_aberration_01.htm

http://www.dpreview.com/learn/?/Glossary/Optical/Vignetting_01.htm

11) Our camera has Minolta GT APO -lens. What does APO mean here?

12) Name one measure of sharpness that can be used to determine if a region in an image is in focus or blurred.

Remember to take **the answers** with you to the lab.

Camera functions

One of the three goals of this exercise is to get us familiar with a system camera. The details of the camera operation will be discussed in the beginning of the exercise. Here are some important settings of a system camera:

In this exercise the use of **Manual Focus** (MF) is required, because the goal is to acquire images with different optical settings. The automatism of the camera will then be partially disabled. The aperture can be selected with **Aperture Priority** -function. The shutter time is then controlled by the camera. The use of a camera stand is required in order to keep the camera accurately on its place between focus adjustments. The DRT-method is sensitive to misalignments between pixels especially when a small window size is used in the determination of the degree of blur.

The light conditions can easily be taken into consideration by the Subject Program function which optimises the camera's performance for some special conditions. These predefined factory settings make the operation of

the camera easy and fast because they help the photographer with the most difficult scenes, such as motion, sunset, moonlight or winter snow that reflects the sunshine.

There are 16 elements in the lens system. Its **focal length** can be adjusted between 7.2mm and 50.8mm. This is the 7x manual **zoom**, $50.8\text{mm} / 7.2\text{mm} = 7.0556$. The image size can be adjusted between 640x480 and 2560x1920 pixels. The TIFF file size will then be between 0.9 and 14.5 MB. For more details about the camera, see this website: http://www.dpreview.com/reviews/specs/Konica_Minolta/minolta_dimage1.asp.

Optical background

We determine the depth of the scene from the degree of blurring. Two differently focused images are convolved with gaussian and laplacian filters. The results are combined to produce an intensity image C , which is mapped to depth information.

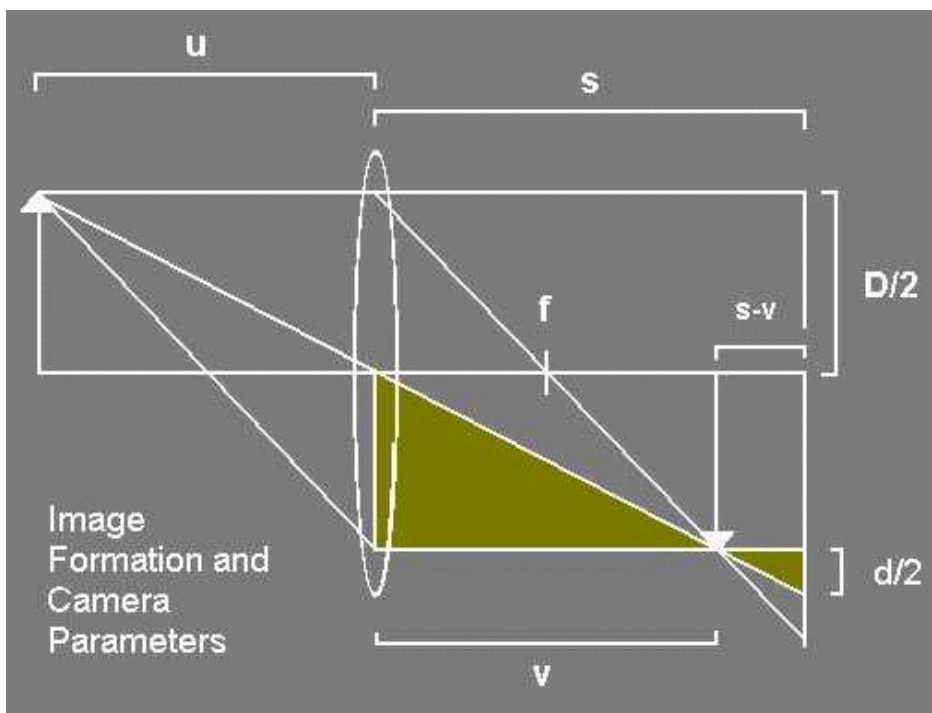


Figure 1: Blur circle radius $d/2$ varies with respect to focal length f and the distances u and s .

The lens law equation

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad (\text{a})$$

and the proportion

$$\frac{D/2}{d/2} = \frac{v}{s-v} \quad (\text{b})$$

are combined and written in Equation (1) with respect to **blur circle diameter** d . It can be determined by aperture diameter D , focal length f , object distance u , proportionality constant k and distance s that is measured from the lens to the image plane. The derivation of Equation (1) is straightforward. We use the triangles in Figure 1 according to Equation (b) and the lens equation (a).

$$d = \frac{D(s-v)}{v} = \frac{D\left(s - \frac{fu}{u-f}\right)}{\frac{fu}{u-f}} = \frac{Duf\left(\frac{s}{f} - \frac{s}{u} - 1\right)}{uf} = Ds\left(\frac{1}{f} - \frac{1}{u} - \frac{1}{s}\right) \quad (1)$$

This means that if the distances v and s are equal,

$$\frac{1}{v} = \left(\frac{1}{f} - \frac{1}{u}\right) = \frac{1}{s} \quad (2)$$

and the blur circle radius is zero. See Figure 1. Hence, the image is in focus. The spread parameter is proportional to blur circle diameter d .

$$\sigma = kd = kDs\left(\frac{1}{f} - \frac{1}{u} - \frac{1}{s}\right) \quad (3)$$

The intensity distribution inside the blur circle is described by the two-dimensional gaussian function.

$$h = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}} \quad (4)$$

The power spectral density function of this gaussian is

$$P(\omega, \nu) = e^{-(\omega^2 + \nu^2)\sigma^2} FF^* \quad (5)$$

This means that blurring is equivalent to exponential decay in the frequency domain. The degree of defocusing is estimated by the difference of the focused image and the defocused image in the frequency domain.

Taking derivative of Equation (5) with respect to the spread parameter we obtain

$$\partial P = -2(\omega^2 + \nu^2)P\sigma\partial\sigma \quad (6)$$

in which

$$\partial\sigma = kD\left(\frac{1}{f} - \frac{1}{u}\right)\partial s \quad (7)$$

that is derived from Equation (3) with respect to s , the distance from the sensor plane.

Using the above equations and defining the relation

$$\sigma\partial\sigma = \frac{-1}{2(\omega^2 + \nu^2)} \frac{\partial P}{P} = C \quad (8)$$

to be a constant with a given set of camera system parameters we obtain:

$$\sigma \partial \sigma = \left(kDs \left(\frac{1}{f} - \frac{1}{u} - \frac{1}{s} \right) \right) \left(kD \left(\frac{1}{f} - \frac{1}{u} \right) \right) \partial s \quad (9)$$

Arranging and replacing

$$k^2 D^2 s \partial s = K \quad (10)$$

the distance to the object can be determined by solving u from:

$$K \left(\frac{1}{f} - \frac{1}{u} - \frac{1}{s} \right) \left(\frac{1}{f} - \frac{1}{u} \right) = C \quad (11)$$

Only one parameter, K , needs to be determined, which makes the calibration of the optical system easy. The value of C is determined by the **Dynamic Referencing Technique** developed by Akihiro Horii at The Royal Institute of Technology, Sweden.

In principle it would be sufficient to compute the power spectral density P and its derivative at a single image point, but to obtain a more robust estimate it is done on a region R (area of which is A) according to:

$$C = -\frac{1}{2A} \iint_R \frac{1}{(\omega^2 + \nu^2)} \frac{\partial P(\omega, \nu)}{P(\omega, \nu)} \partial \omega \partial \nu \quad (12)$$

C is therefore only an average over an area of the image. Compromises have to be made between the resolution and the smoothness of the depth recovery when selecting the size of the area.

The Fourier power is needed for computing C . We can make use of Parseval's theorem. It states that the Fourier power (the integral of the squared Fourier components of the signal) can be obtained from the squared values over the spatial domain. More specifically,

$$\begin{aligned}
\frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |F(\omega, \nu)|^2 \partial\omega \partial\nu &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x, y)|^2 \partial x \partial y \\
&= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P(\omega, \nu) \partial\omega \partial\nu
\end{aligned} \tag{13}$$

in which $P(\omega, \nu)$ is the power spectrum. This means computing the original signal power over a range of frequencies by utilizing the integral of the square of the filtered signal. By filtering the image with a band-pass filter (laplacian filter) the signal is restricted to a limited range of frequencies.

To compute C , the value of $(\omega^2 + \nu^2)$ in Equation (8) has to be determined. But the term depends on the texture of the image. Moreover, the images have to contain texture. Otherwise there would not be blurring to be analysed. The problem with the dependence on the texture is avoided by the Dynamic Referencing Technique which uses "dummy blurring" obtained by gaussian convolution. The power difference between the two images is

$$\frac{\partial P}{P} = \frac{2(P_2 - P_1)}{P_2 + P_1} \tag{14}$$

and the power difference between the first image and the dummy gaussian blurred image is

$$\frac{\partial P_g}{P_g} = \frac{2(P_g - P_1)}{P_g + P_1} \tag{15}$$

The value of C can be expressed by the two constants

$$C_1 = -\frac{1}{2(\omega^2 + \nu^2)} \frac{\partial P}{P} \tag{16}$$

and

$$C_2 = -\frac{1}{2(\omega^2 + \nu^2)} \frac{\partial P_g}{P_g} \quad (17)$$

the ratio of which is the degree of blurring, the relative distance difference. This results in the compact expression of C which is approximated by the algorithm.

$$C = \frac{C_1}{C_2} = \frac{P_g}{\partial P_g} \frac{\partial P}{P} \quad (18)$$

Dynamic Referencing Technique

- 1) Two greyscale images with different camera settings (focus distances) are filtered with a gaussian filter G .
- 2) The resulting images $G1$ and $G2$ are bandpassed by laplacian filtering. The laplacian energies $L1$ and $L2$ are obtained by squaring the pixel values after filtering. These energies are summed over an area A to produce $P1$ and $P2$.
- 3) The relative power difference $PD1$ is calculated as $2(P2-P1)/(P2+P1)$.
- 4) The "dummy" blurred image $P1g$ is obtained by repeating gaussian and laplacian filtering steps 1) and 2) **only** for $G1$.
- 5) The relative power difference $PD2$ between the original image and $P1g$ is calculated as $2(P1g-P1)/(P1+P1g)$.
- 6) C is calculated from $PD1/ PD2$. (Equation 18)
- 7) Distances are solved with the calibrated K and $s=1/(1/f-1/q)$ which is computed from q (the focused distance that corresponds to ν in Figure 1)

Depth analysis with Matlab

- 1) Create the filters: 2-D gaussian low-pass filter G of size 3×3 and standard deviation of 0.85 pixels, 2-D laplacian band-pass filter $L = \begin{bmatrix} -2 & -1 & -2 \\ -1 & 12 & -1 \\ -2 & -1 & -2 \end{bmatrix}$. Examine their **frequency responses** to get an idea of what the algorithm actually does to the images.
- 2) Apply the DRT-algorithm to a normalized greyscale image pair. Use descriptive variable names in order to make it easier to interpret your code. Choose the operating window (block-processing) size according to the size of the target image (the compromise). Take a look at the intermediate images to deepen your knowledge of the process.
- 3) Cut away the edges of the resulting matrix C so that the 3-D plot of the scene is not affected by the side-effects of filtering. This way the resulting surface looks more illustrative.
- 4) Smoothen (average filter) the coefficient image C reasonably with respect to image size and scene type.
Note that the "right" **window size** is determined by the particular application and the level of detail recovery requirements of the scene. If the window is too small, the result is not obtained. The algorithm requires an area. If the window size is too big, the details are lost. This is critical. If the difference in the **amount of defocusing** between the two images is too small, try e.g. images #1 and #4 instead of images #1 and #2. If there is too much defocusing the algorithm does not work well.
- 5) Convert the coefficient values C into depth values using information from the experimental conditions. Use Equation (11) to calculate the actual depth values for each **image area**.
- 6) The order of the two chosen images in the DRT-algorithm affects the "polarity" of C . For example, try using images #1 and #4 instead of images #4 and #1. This is important when we want to make the 3-D plot have an illustrative pose. The goal is to illustrate the scene with information about its dimensions: the surface should match the original view as well as possible.

Useful tips

- 1) Recall the difference in Matlab's commands with matrices:
 $\mathbf{A}.\mathbf{B}$ versus $\mathbf{A}/\mathbf{B}$.
- 2) Normalize the images with respect to the average intensity of each image before applying the algorithm.
- 3) There exists a variety of commands for controlling the appearance of a Matlab plot. The most important goal is that the depth values correspond to the real scene at the z-axis.
- 4) Substitution of $\mathbf{X}=(\mathbf{1}/\mathbf{f}-\mathbf{1}/\mathbf{u})$ into Equation (11) makes it easy to obtain the solution. Calibration of K with respect to the scene depth is illustrated in Figure 2. Here the focused distance ($C=0$) is 80cm.

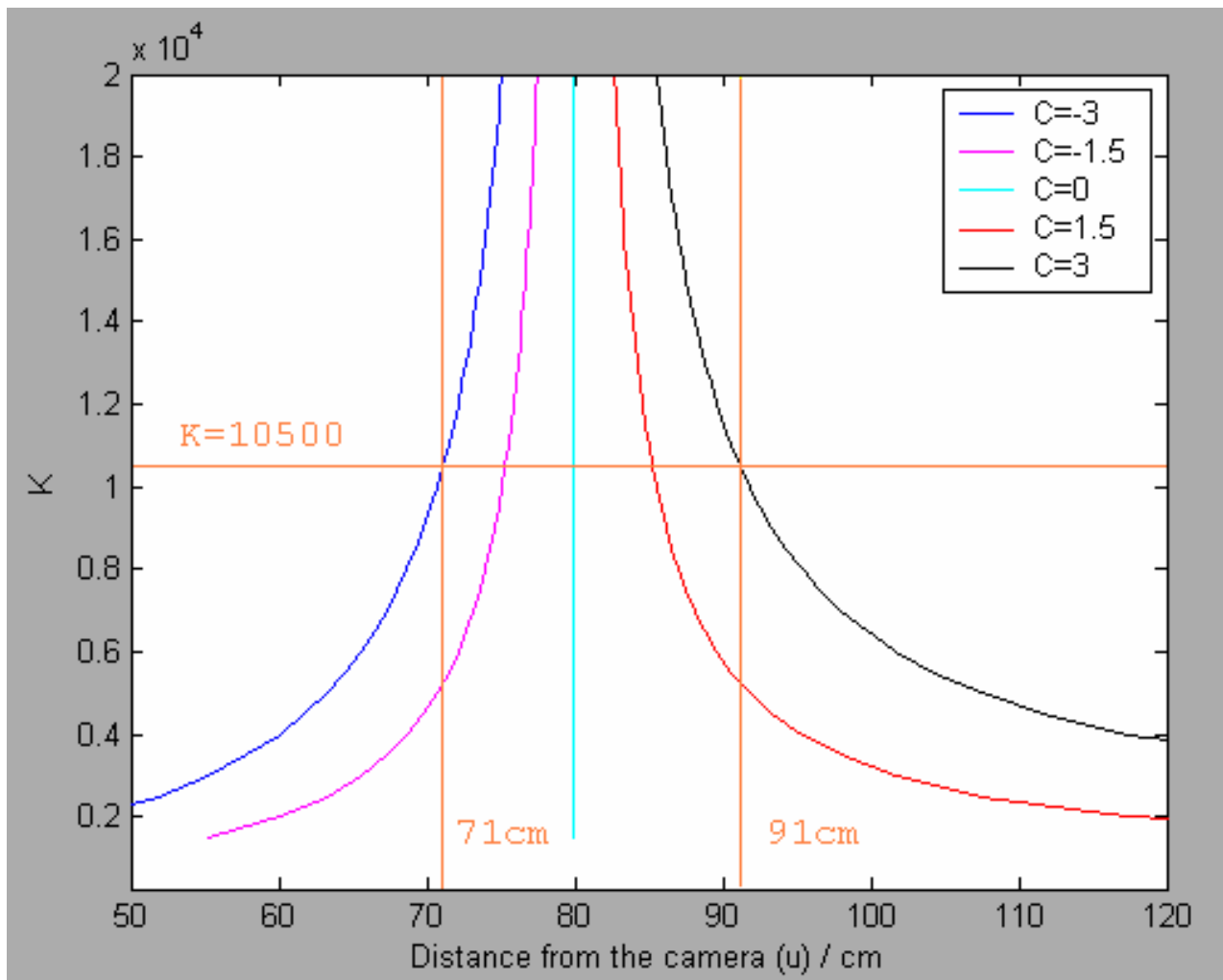


Figure 2: Parameter K with respect to values of matrix C . The distance difference (here 20cm) determines the value of K .

- 5) Some handy functions to be used in this exercise are
 [rgb2gray flipdim fspecial filter2 blkproc roots surf text].
- 6) The following example creates a surface and shows how to adjust its properties:
- ```
figure(123); newsurface=surf(depthdata)
get(newsurface)
[displays the alternatives for adjusting the parameters]
get(axes) [displays information about the axes]
set(newsurface,'Facecolor','interp','Facelighting','none', ...
'Linestyle','none')
figure(123); set(gca,'Zdir','reverse')
[activates the figure123 and changes the z-axis data direction]
zdata=get(newsurface,'Zdata') [obtains data from the plot]
help graph3d [displays a list of colormaps]
colormap hot [chooses a color scheme "hot"]
whitebg [changes the background color]
```
- 7) Here is an example of computing the actual distances for a given set of  $s, f$  and  $K$  based on matrix  $C$  that contains the relative distances of the scene:

```
scl=size(C,1); sc2=size(C,2);
distances=ones(size(C));
for verind=1:scl
 for horind=1:sc2 % first solve X using equation(10) ...
 X=roots([first(x^2)coefficient ...
 second(x^1)coefficient ...
 third(x^0)coefficient]);
 if (X(1)~=0)
 distances(verind,horind)= % then solve u (using X(1) and f)
 else
 distances(verind,horind)= % then solve u (using X(2) and f)
 end
 end
end
end
```

# Requirements

## 1) Answering the **preliminary questions**.

This will be checked before taking the photos.

## 2) Getting familiar with the camera system and **image acquisition**.

## 3) The **m-file** that is and error-free and produces the required figure with the scene's depth information by calling the filename.

Copies of the original images are saved on the laboratory computer for checking the results of each group.

Use exactly the same image names in your Matlab code for the images that you acquired in the lab.

The grade is given based on answering the pre-questions, the appearance of the m-file and the result. There are two categories:

**3 points:** Accepted answers to the questions, successful acquisition of the relative depth data **matrix C**, a plot like on the first page of this exercise. The real **distances** [cm] are acquired. The **distance difference** between the farthest and closest objects and the **depth values** correspond to the real scene. This is sufficient for passing this exercise.

**5 points:** In addition to the previous category, matrix C is represented as a **3-D plot**, a surface that has an informative pose (one that corresponds to the original scene) with informative **colours**. Red plot area indicates that the distance is small, thus the object is close to the camera. Blue means the opposite. The **Matlab code** is clear and **informative**. There are **comments** within the m-file that explain the code. **Variable names** are descriptive. The figure has a **title** and the **z-axis label** [cm] can be seen in the plot.

Return to hirvone2@cs.tut.fi by **17.3.2006**.